List of Programs

1. Write a JavaScript program to check whether the given number is perfect, abundant, or deficient. Use alert box to display the output.

2. Design a JavaScript program to display the multiplication table by accepting the number and limit.

3. Write a Java Script program to store different colors in an array and change the background color of the page using these array elements.

4. Write a Java Script program to change the text color and back color of a Textbox using on focus and On Blur event.

5. Write a Java Script function to display current date and time.

6. Write a Java Script program that calculates the squares and cubes of the numbers 0 to 10 and outputs HTML texts that displays the resulting values in a HTML table format.

7. Design a student registration form and apply validation on it by using external Java Script.

8. Design a simple calculator by using html, css and Java script.

9. Write a Java Script program to calculate factorial of any number by using html.

10. Write a JavaScript function to check whether a given value is IP value or not

```html
<!-- Program 1: Write a JavaScript program to check whether the given
   number is Perfect, Abundant, or Deficient. -->

<!DOCTYPE html>
<html>
<body>
<script>
let num = parseInt(prompt("Enter a number:"));
let sum = 0;

for (let i = 1; i <= num / 2; i++) {
    if (num % i === 0) sum += i;
}

if (sum === nu)
{
    alert("Perfect Number");
}
else if (sum > num)
{
    alert("Abundant Number");
}
else
{
    alert("Deficient Number");
}
</script>
</body>
</html>
```

```html
<!-- Program 2: Design a JavaScript program to display the
multiplication table by accepting the number and limit. -->

<!DOCTYPE html>
<html>
<body>
<input type="number" id="num" placeholder="Number">
<input type="number" id="limit" placeholder="Limit">
<div id="result"></div>
<button onclick="table()">Show Table</button>


<script>
function table() {
    let n = parseInt(document.getElementById("num").value);
    let l = parseInt(document.getElementById("limit").value);
    let out = "";

    for (let i = 1; i <= l; i++) {
        out += `${n} x ${i} = ${n*i}<br>`;
    }
    document.getElementById("result").innerHTML = out;
}
</script>
</body>
</html>
```

```html
<!-- Program 3: Write a JavaScript program to store different colors
in an array and change the background color of the page. -->

<!DOCTYPE html>
<html>
<body>
<button onclick="changeColor()">Change Background</button>

<script>
let colors = ["red", "green", "blue", "yellow", "pink"];
let i = 0;

function changeColor() {
    document.body.style.backgroundColor = colors[i];
    i = (i + 1) % colors.length;
}
</script>
</body>
</html>
```

```html
<!-- Program 4: Write a JavaScript program to change the text color
and background color of a textbox using onfocus and onblur events. -->

<!DOCTYPE html>
<html>
<body>
<input type="text"
onfocus="focusFun(this)"
onblur="blurFun(this)"
placeholder="Enter text">

<script>
function focusFun(x) {
    x.style.background = "yellow";
    x.style.color = "red";
}
function blurFun(x) {
    x.style.background = "white";
    x.style.color = "black";
}
</script>
</body>
</html>
```

```html
<!-- Program 5: Write a JavaScript function to display
the current date and time. -->

<!DOCTYPE html>
<html>
<body>
<button onclick="showDate()">Show Date & Time</button>
<p id="dt"></p>

<script>
function showDate() {
    let now = new Date();
    document.getElementById("dt").innerHTML = now;
}
</script>
</body>
</html>
```

```html
<!-- Program 6: Write a JavaScript program to calculate
the squares and cubes of numbers from 0 to 10. -->

<!DOCTYPE html>
<html>
<body>
<table border="1">
<tr>
<th>Number</th>
<th>Square</th>
<th>Cube</th>
</tr>

<script>
for (let i = 0; i <= 10; i++) {
    document.write("<tr>");
    document.write("<td>" + i + "</td>");
    document.write("<td>" + (i*i) + "</td>");
    document.write("<td>" + (i*i*i) + "</td>");
    document.write("</tr>");
}
</script>
</table>
</body>
</html>
```

```
<!-- Program 7: Design a student registration form
and apply validation using external JavaScript. -->

<form onsubmit="return validate()">
Name: <input type="text" id="name"><br>
Email: <input type="text" id="email"><br>
Age: <input type="number" id="age"><br>
<button type="submit">Submit</button>
</form>

<script src="validate.js"></script>// Validation script for Program 7
// name javascript file as validate.js
function validate() {
    let name = document.getElementById("name").value;
    let email = document.getElementById("email").value;
    let age = document.getElementById("age").value;


    if (name === "") {
        alert("Name required");
        return false;
    }
    if (!email.includes("@")) {
    alert("Invalid email");
    return false;
  }
  if (age < 18) {
    alert("Age must be 18+");
    return false;
  }
  alert("Form submitted successfully");
  return true;
}
```

```html
<!-- Program 8: Design a simple calculator using
HTML, CSS and JavaScript. -->

<!DOCTYPE html>
<html>
<body>
<input id="a" type="number">
<input id="b" type="number"><br>

<button onclick="calc('+')">+</button>
<button onclick="calc('-')">-</button>
<button onclick="calc('*')">*</button>
<button onclick="calc('/')">/</button>

<p id="res"></p>

<script>
function calc(op) {
    let x = parseFloat(a.value);
    let y = parseFloat(b.value);
    let r;

    if (op === '+') r = x + y;
    if (op === '-') r = x - y;
    if (op === '*') r = x * y;
    if (op === '/') r = x / y;

    res.innerHTML = "Result: " + r;
}
</script>
</body>
</html>
```

```html
<!-- Program 9: Write a JavaScript program to
calculate factorial of a number using HTML. -->

<!DOCTYPE html>
<html>
<body>
<input type="number" id="n">
<button onclick="fact()">Factorial</button>
<p id="ans"></p>

<script>
function fact() {
    let num = parseInt(n.value);
    let f = 1;
    for (let i = 1; i <= num; i++) {
        f *= i;
    }
    ans.innerHTML = "Factorial = " + f;
}
</script>
</body>
</html>
```

```html
<!-- Program 10: Write a JavaScript function to
check whether a given value is a valid IP address. -->

<!DOCTYPE html>
<html>
<body>
<input type="text" id="ip" placeholder="Enter IP">
<button onclick="checkIP()">Check</button>
<p id="out"></p>

<script>
function checkIP() {
    let ip = document.getElementById("ip").value;
    let regex = /^(\d{1,3}\.){3}\d{1,3}$/;

    if (regex.test(ip)) {
        out.innerHTML = "Valid IP Address";
    } else {
        out.innerHTML = "Invalid IP Address";
    }
}
</script>
</body>
</html>
```